Spectrally–Spatially Coupled Dynamic Reduction for Efficient Hyperspectral Image Super-Resolution

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***Abstract*—Hyperspectral image super-resolution faces a fundamental challenge in practical applications: achieving high-fidelity reconstruction on resource-constrained edge devices while maintaining manageable computational complexity. Although recent deep learning-based single-image methods have shown remarkable success, their substantial computational overhead hinders deployment. Meanwhile, emerging architectures like Mamba offer efficient long-range modeling potential but struggle with the inherently high dimensionality of hyperspectral data, leading to prohibitively long sequences and excessive computational loads. Existing lightweight strategies often process spectral and spatial dimensions independently, failing to exploit their intrinsic coupled correlation and resulting in a suboptimal performance-efficiency tradeoff. To address this, this article proposes a spectrally–spatially coupled dynamic lightweight network (S^2CD-Net). Its core innovation is a novel paradigm of coupled reduction and independent learning. We first design a dual-stage collaborative dimensionality reduction mechanism, where spectral selection and spatial compression mutually guide and refine each other in a closed loop, enabling intelligent and precise removal of cross-dimensional redundancy. Subsequently, the resulting low-redundancy spectral and spatial sequences are fed into dedicated, lightweight SS2D scanners for independent and in-depth global modeling, ensuring rich feature representation while reducing complexity. Experiments demonstrate that our method requires only minimal parameters and computational cost, yet achieves reconstruction quality comparable to and even surpassing current nonlightweight state-of-the-art methods, establishing a new optimal balance between performance and efficiency. The code is available at https://github.com/XYLGroup/S2CD-Net**

***Index Terms*—Hyperspectral image, lightweight, SS2D, super-resolution.**

I. Introduction

HYPERSPECTRAL imaging captures fine spectral signatures across hundreds of contiguous bands, enabling detailed material analysis and showing significant value in environmental monitoring [1], [2], precision agriculture [3], [4], and resource exploration [5], [6]. However, constrained by sensor physics, hyperspectral images typically suffer from low spatial resolution, limiting their analytical utility in practical applications. Hyperspectral image super-resolution aims to reconstruct high-spatial-resolution cubes from low-resolution (LR) inputs, whose core scientific challenge lies in effectively balancing the inherent tradeoff between spectral fidelity and spatial detail enhancement [7], [8], [9]. When deployed on resource-limited edge devices such as satellites and drones, this tradeoff manifests as a direct tension between reconstruction accuracy and computational cost, thereby creating an urgent demand for lightweight and adaptive algorithm design.

In recent years, data-driven methods-based on deep learning [10], [11], [12], [13], [14], [15], [16], [17] have significantly advanced the reconstruction accuracy of HSI-SR. These methods are primarily divided into two paradigms. The fusion-based paradigm [18], [19], [20], [21] incorporates auxiliary high-resolution (HR) images but faces strict co-registration requirements and limited data availability, hindering its practical deployment. The single-image-based paradigm [10], [11], [22], [23], [24], which relies solely on the input LR-HSI, offers greater generality. Architectures such as CNNs [25], [26], [27] and transformers [16], [28] have pushed the performance boundary, yet they often entail substantial computational overhead, making them unsuitable for edge devices. Recent lightweight attempts, including static pruning or fixed sampling, often treat the spectral and spatial dimensions independently, failing to exploit their intrinsic coupled redundancy and leading to suboptimal performance-complexity tradeoffs. Concurrently, state-space models, particularly Mamba, have emerged as a promising backbone due to their linear complexity in modeling long-range dependencies. However, applying Mamba [29], [30], [31] directly to HSI-SR is nontrivial, as the raw hyperspectral cube yields an excessively long sequence, resulting in prohibitive computational and memory costs. Therefore, a critical challenge emerges: leveraging the global modeling strength of models like Mamba while radically reducing the dimensionality of the data they must process. This necessitates a fundamental rethinking of front-end dimensionality reduction, moving beyond independent operations to a collaboratively designed mechanism that respects the spectral–spatial coupling inherent in the data.

To address the above challenges, this article proposes a spectrally–spatially coupled dynamic lightweight network, called S^2CD-Net. Its core innovation is a dual-stage dimensionality reduction mechanism that forms a collaborative

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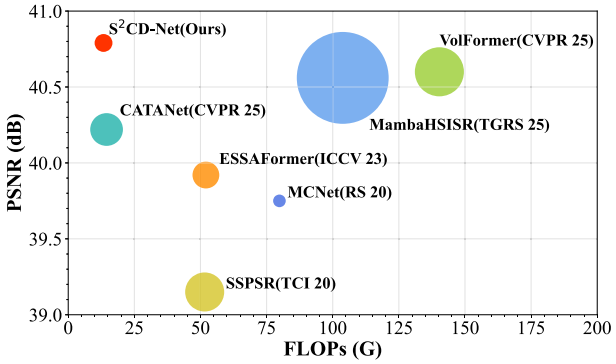


Fig. 1. All competing methods are evaluated on three metrics: PSNR (vertical axis), FLOPs (horizontal axis), and model size (circle radius). S^{2}CDNet outperforms other SOTA methods while requiring less computation.

optimization loop between spectral and spatial compression. In the first stage, an adaptive spectral selection and compression module dynamically selects informative bands, guided not only by their own importance but also implicitly by spatial context. In the second stage, a semantic-aware spatial merging module merges spatially correlated pixels based on spectral similarity and reorganizes them via semantic ordering. Crucially, these two stages are interlinked: the refined spectral subset provides a purer basis for spatial similarity computation, while the aggregated spatial semantics offer feedback to refine subsequent spectral selection. This design embodies a clear philosophy of coupled reduction and independent learning. The coupled reduction stage performs intelligent, synergistic compression to produce low-redundancy spectral and spatial sequences. These sequences are then processed by dedicated, lightweight SS2D scanners that perform independent, in-depth global modeling within each dimension. This separation ensures that the richness of information is preserved and enhanced during learning, while the preceding coupled reduction guarantees that the learning is performed on a computationally tractable and semantically coherent input. Consequently, our approach enables precise removal of redundancy while maximally retaining critical information for high-fidelity reconstruction.

Experimental results and Fig. 1 demonstrate that the proposed method achieves reconstruction quality comparable to, and in some cases surpassing, current non-lightweight state-of-the-art methods, while requiring minimal parameters and computational cost. This successfully balances high performance with low complexity. The main contributions of this work are summarized as follows.

1) We propose a novel spectrally–spatially coupled dimensionality reduction paradigm. It introduces a two-stage collaborative mechanism in which spectral and spatial compression iteratively steer and optimize one another, enabling intelligent and accurate elimination of cross-dimensional redundancy in a lightweight HSI-SR model for the first time.
2) We design a network-guided adaptive band rewriting mechanism and a semantic-aware token merging strategy. In the spectral dimension, a network-based adaptive Markov Chain Monte Carlo (NA-MCMC) sampling framework is proposed for dynamic band selection. In the spatial dimension, a spectral-similarity-based semantic-aware mapping is constructed for intelligent pixel merging.
3) We provide comprehensive experimental validation. Our method sets a new state-of-the-art tradeoff between performance and efficiency on multiple public datasets, demonstrating its superiority and practical potential for edge deployment.

II. Related Work

A. HSI Super-Resolution Methods

Requiring no auxiliary data, single-image SR methods are more practical and flexible than fusion-based approaches. Early research predominantly relied on model-driven approaches and prior knowledge constraints, such as sparse coding methods [32], [33], low-rank representation methods [34], [35], and variational inference methods [36], [37]. Although these methods offer interpretability, they suffer from high computational complexity and weak generalization of hand-crafted priors.

Deep learning has fundamentally revolutionized the data-driven paradigm for HSI-SR, initiating a shift from hand-crafted priors to data-driven representations [38]. Among early 3D-CNN [27] architectures, 3D-FCNN pioneered the fully convolutional design, achieving end-to-end mapping through a combination of mirror padding and fully convolutional layers, which fundamentally eliminates blocking artifacts and boundary luminance discontinuities. MCNet [10] further transcends the limitations of single-scale features by constructing a parallel multibranch 3-D convolutional structure to capture features at diverse receptive fields. These methods significantly improve accuracy compared to band-independent processing; however, the number of parameters in 3-D convolutions grows linearly with the spectral dimension. To mitigate gradient vanishing via residual connections, SSPSR [11] integrates spatial and spectral prior constraints to enhance feature discriminability, employs grouped convolution with progressive upsampling, and fuses multiscale contextual information through a multibranch architecture.

Attention mechanisms revolutionarily introduce dynamic weight recalibration, enabling networks to adaptively focus on critical features. AD-DFGNet [39] explicitly models high-frequency structures via gradient-texture attention and feature aggregation to overcome implicit learning's limitations in gradient-texture separation, enhancing edge fidelity. However, local convolutions constrain long-range spectral–spatial dependencies, motivating a shift to transformers. MSDformer [16] employs multiscale deformable transformer modules to enable attention windows to dynamically adjust sampling positions based on image content, focusing on key structural regions. DSSTSR [14] adapts Swin transformer to hyperspectral domain, leveraging the window shifting mechanism to achieve local–global collaborative modeling and capture long-range dependencies. Despite their global modeling potential,

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quadratic complexity and small-sample overfitting limit transformers for hyperspectral imagery.

Recently emerged state-space models provide a linear-complexity alternative, with the Mamba architecture demonstrating great potential in HSI-SR through its selective scanning mechanism. MambaHSISR [29] first introduces Mamba to hyperspectral super-resolution, designing dual spatial–spectral Mamba subnetworks that collaboratively model long-range dependencies via spatial scanning and spectral correlation modules. However, Mamba's causal nature restricts each token to depend solely on predecessors, limiting spatial context interaction [40], while its extensive hidden states readily induce channel redundancy.

B. Lightweight Strategies

Existing lightweight strategies can be broadly categorized into three groups: posttraining model compression, inference-time lightweight techniques, and efficient architecture design, which aim to streamline pretrained models, perform dynamic optimization during execution, and construct efficient networks from the ground up, respectively.

1) Posttraining Model Compression: This category streamlines large models after their training is complete, aiming to reduce their storage and computational costs for deployment without altering the core network architecture. Knowledge distillation [41], [42], [43], [44], achieves efficient compression by transferring knowledge from a teacher model to a student model. EigenSR [41] employs eigenimages as an intermediary to distill powerful spatial priors from pretrained RGB models into the HSI-SR task, enabling a lightweight student network to achieve superior performance under data-scarce conditions. Parameter pruning [45], [46], [47] compresses models by removing redundant weights or structures. LKCA-Net [47] exploits the low-rank nature of upsampling layers for model compression via structured pruning. While these post hoc methods offer direct compression benefits, their effectiveness is bound by the original model's capacity. The process requires careful design to minimize performance loss and often involves additional fine-tuning steps.

2) Inference-Time Lightweight Techniques: This category focuses on dynamically optimizing computational pathways during inference based on the input data, with the core objective of achieving "computation on demand" [48]. Selective feature processing [49], [50] enables the network to dynamically focus on critical information. SRLF-Net [51] proposes a selective relearning mechanism to optimize channel redundancy, dynamically filtering key spectral channels for relearning through gating units to suppress invalid information propagation. EGSRN [52] proposed an edge-guided lightweight convolution module that utilizes Sobel operators to extract edge priors and dynamically modulate feature responses. However, these dynamic techniques face key challenges. First, they reduce interpretability by making the decision process opaque and hard to trace. Second, they increase design complexity, requiring specialized algorithms and extra training overhead.

3) Efficient Architecture Design: This category achieves lightweighting from the ground up in network design, including designing lightweight modules [52], [53], adopting linear-complexity architectures [54], [55], [56], and implementing parameter sharing [57], [58]. Notably, Guo et al. [53] embedded spectral mixing priors into a five-layer microautoencoder to achieve extreme compression, innovatively injecting physical interpretability into network design. In recent years, to address the quadratic complexity of transformer architectures, research has focused on dynamic mechanisms and linear architectures. ESSAformer [54] achieves linear complexity via spectral correlation coefficient (CC) kernelized self-attention, using a squared exponential kernel function to reorder multiplication and reduce complexity from $O(N^2)$ to $O(N)$. However, these methods still exhibit limitations in the performance-efficiency tradeoff and dynamic adaptability during dimensionality reduction, constraining their deployment and practical application on edge devices.

C. SGN Strategy

The semantic guided neighboring (SGN) strategy [59] reorders tokens based on semantic similarity, which makes pixels that are spatially distant but semantically similar adjacent in the sequence space, thereby enhancing the information interaction capability during the sequential scanning process.

Given a feature representation $\mathbf{X} \in \mathbb{R}^{HW\times C}$, semantic assignment probabilities are first obtained as

$$\mathbf{\Pi} = \text{Softmax}\,(\mathbf{X}\mathbf{W}_s) \tag{1}$$

where $\mathbf{W}_s$ is a learnable weight matrix projecting each token's feature to a semantic space for computing semantic assignment probabilities. Followed by approximate discrete semantic labels using Gumbel-Softmax

$$c_n = \arg\max_k \text{GumbelSoftmax}\,(\mathbf{\Pi})_{n,k}\,. \tag{2}$$

Tokens are then reordered according to the semantic labels

$$\mathbf{X}_{seq} = \mathbf{X}_\pi, \quad \pi = \text{SortIndex}\,(\{c_n\}) \tag{3}$$

and mapped back to the spatial layout after sequential scanning. This semantic-driven reordering shortens the interaction distance between related tokens in the sequence space and alleviates the long-range decay problem in sequential modeling.

III. Proposed Method

This section introduces the S^{2}CD-Net two-stage architecture designed for lightweight HSI-SR. We first describe the notation and overall framework, then present the two key components separately: the S2C and semantic-aware merging (S2M) module.

A. Preliminary

In this article, vectors, matrices, and tensors are denoted by boldface letters. Superscripts indicate iteration indices, while subscripts convey the semantic meaning of variables. Regarding data dimensions, $\mathbb{R}^{H\times W}$ represents the spatial size of high-dimensional data, and $\mathbb{R}^{h\times w}$ that of low-dimensional data, where H, h are the heights and W, w are the widths; C denotes the high-dimensional channel number, and c the

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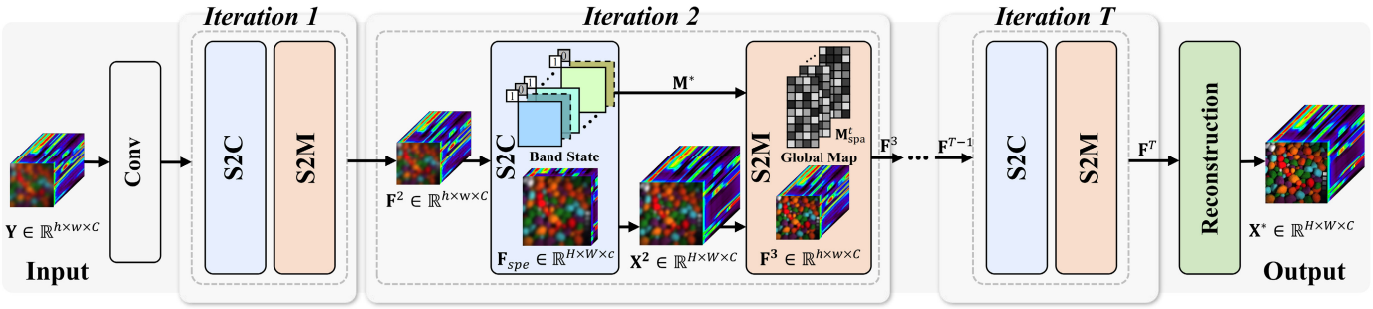


Fig. 2. Overall framework of the proposed S^{2}CD-Net. The process extracts shallow features through a convolutional layer, then uses lightweight iterative optimization with S2C and S2M to refine deep features, before final upsampling for reconstruction.

low-dimensional channel number. $\odot$ denotes element-wise multiplication, and $\times$ represents matrix multiplication. $\in$ indicates element membership in a set, $\sim$ denotes that a variable follows a probability distribution, and $\subset$ indicates a proper subset relation between sets. In the S2C module, the band selection state is defined as a vector $\mathbf{m} \in \{0,1\}^C$, where $m_i = 1$ indicates that the ith band is selected and $m_i = 0$ indicates it is discarded. $\mathcal{I}$ represents the sampled band subset, consisting of the indices i of the selected bands. Inside the module, L batch-sampling cycles are performed, which are independent of the T outer iterations of the overall framework. In the S2M module, i and j denote the row and column indices of the matrix, respectively. $\|\cdot\|$ denotes the norm of a vector.

B. Overall Framework

The overall architecture of the proposed S^{2}CD-Net is illustrated in Fig. 2. The framework achieves joint optimization of spectral and spatial dimensionality reduction through alternating optimization between the S2C module and the S2M module, and finally generates a HR HSI via the reconstruction module.

The input LR-HSI $\mathbf{Y}$ first undergoes shallow feature extraction via a convolutional layer

$$\mathbf{F}^1 = \mathrm{Conv}(\mathbf{Y}). \tag{4}$$

Subsequently, the feature is fed into an iterative deep feature extraction process involving dual modules. For the spectral dimension reduction optimization, this is achieved via the S2C module

$$\mathbf{X}^t, \mathbf{m}^* = \mathrm{S2C}(\mathbf{F}^t) \tag{5}$$

where S2C$(\cdot)$ denotes the spectral selection and compression module. The band selection state $\mathbf{m}^*$ and the output feature $\mathbf{X}^t$ are then fed into the subsequent S2M module

$$\mathbf{F}^{t+1} = \mathrm{S2M}(\mathbf{X}^t, \mathbf{m}^*) \tag{6}$$

where S2M$(\cdot)$ denotes the spatial S2M module, and the $\mathbf{F}^{t+1}$. Furthermore, the output from the current tth iteration is fed back into the S2C module for the next optimization round. After T iterations, the final output features $\mathbf{F}^T$ is then reconstructed into the HR-HSI

$$\mathbf{X}^* = \mathbf{H}_{rc}(\mathbf{F}^T) \tag{7}$$

where $\mathbf{H}_{rc}(\cdot)$ is the reconstruction network. The details of this iterative optimization are provided below.

Algorithm 1 Overall Framework of S2C

Input: $\mathbf{F}^t \in \mathbb{R}^{h\times w\times C}$, $\mathbf{m}^1 = \{m_i{=}1\}_{i=1}^C$
$\mathbf{F}_u = \mathrm{Upsampling}(\mathbf{F}^t)$
% Band Importance Evaluation
$\mathbf{s} = \mathrm{sigmoid}(\mathbf{W}_2 \cdot \mathrm{GELU}(\mathbf{W}_1 \cdot \mathrm{GAP}(\mathbf{F}_u)))$
% Adaptive Band Selection
construct the adaptive proposal distribution via (11)
for $l \leftarrow 1, 2, \ldots, L$ **do**
  randomly sample band subset $\mathcal{I}^l$ via (12)
  select scores s_i, states m_i^l for corresponding band i
  compute rewriting factor γ_i via (13)
  sample random thresholds $\epsilon_i \sim \mathrm{Uniform}(0,1)$
  get updated band selection state $\mathbf{m}^{l+1}$ via (14)
end
$\mathbf{F}_d = \mathrm{Conv}(\mathbf{F}_u \odot \mathbf{m}^*)$
% Feature Refinement
$\mathbf{F}_{spe} = \mathrm{Sort\text{-}SS2D}(\mathrm{LN}(\mathbf{F}_d))$
$\mathbf{X}^t = \mathrm{MLP}(\mathrm{LN}(\mathrm{Concat}(\mathbf{F}_d, \mathbf{F}_{spe}))) + \mathbf{F}_u$
Output: $\mathbf{X}^t, \mathbf{m}^*$

C. Spectral Selection and Compression Module

As described in Section III-B, the S2C module is designed to achieve dynamic lightweight processing in the spectral dimension. We model the band selection process as an NA-MCMC sampling framework. The core idea is to leverage a deep network to learn band importance priors, transform band selection into a binary state iterative sampling process of MCMC, and enable the iterative process to converge to the optimal band subset with maximum information content and minimum redundancy by adaptively adjusting the sampling distribution and rewriting mechanism. This framework not only retains the global search characteristics of MCMC sampling but also breaks through the limitation of fixed distributions in classical MCMC via network learning, realizing data-driven adaptive optimization [60], [61]. Algorithm 1 and Fig. 3(a) outline the overall workflow.

1) Band Importance Evaluation: We design a lightweight network $\Phi(\cdot)$ to quantify the effective information of each band. The network takes the upsampled feature $\mathbf{F}_u \in \mathbb{R}^{H\times W\times C}$ as input and outputs the band importance score vector $\mathbf{s} = \Phi(\mathbf{F}_u) \in \mathbb{R}^C$.

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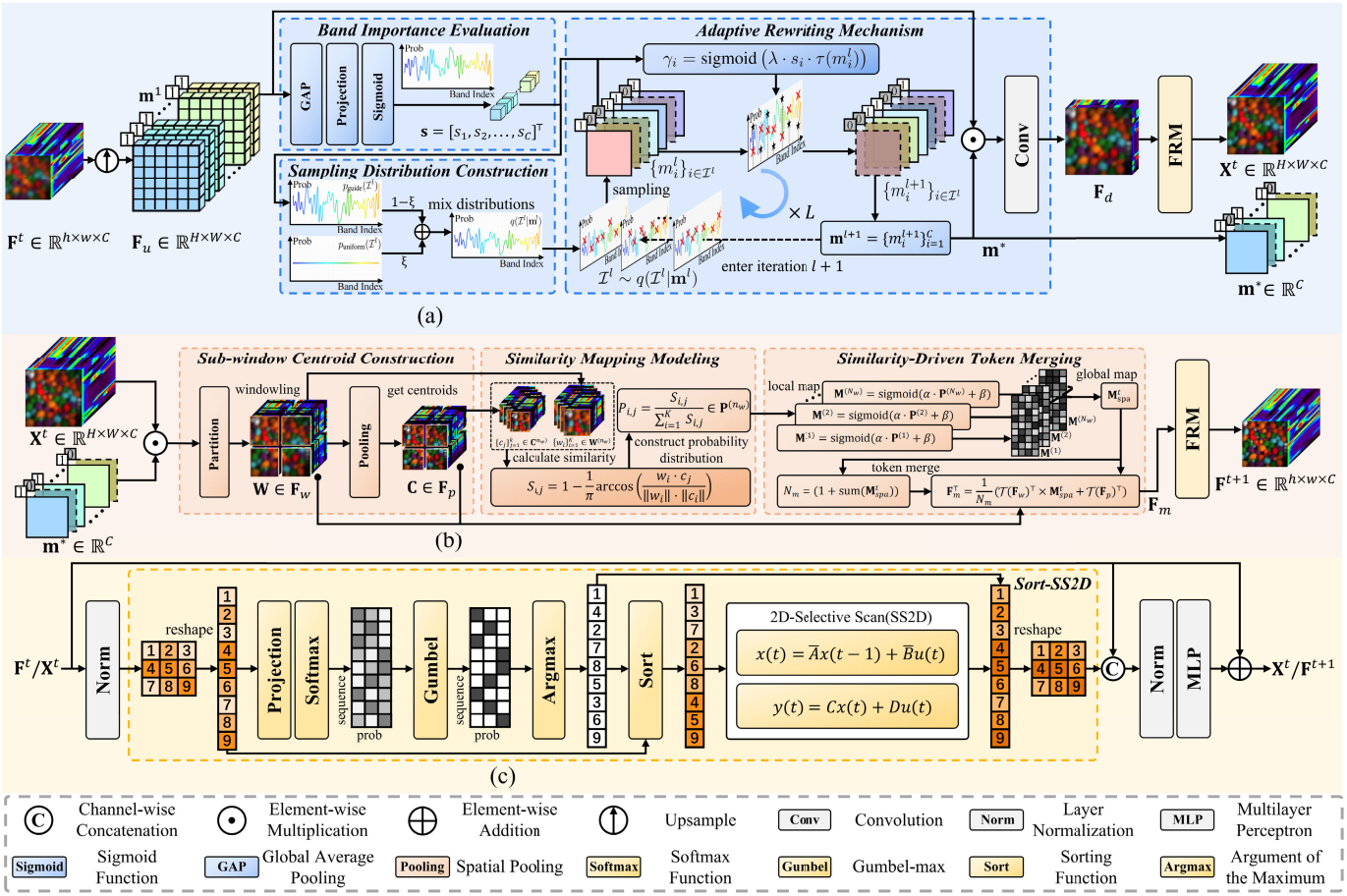


Fig. 3. Overall architecture of the key modules in S^{2}CD-Net. (a) Upsampled features construct mixture distributions based on global band scores, where the band-score curves are derived from the Chikusei $\times 2$ super-resolution task. Adaptive channel selection then employs iterative band-subset sampling and state rewriting. (b) Selected bands are followed by windowing and pooling; similarity maps between pooled centroids and window pixels are then merged into a global map to achieve spatial reduction. (c) Redundancy-removed features are then refined via unidirectional SS2D scanning and an MLP layer.

Specially, $\Phi(\cdot)$ first applies global average pooling (GAP) to aggregate information into a channel-wise descriptor vector. This vector is then nonlinearly transformed by a two-layer feed-forward network (with a GELU activation function in between). The entire process is formulated as

$$\mathbf{s} = \text{sigmoid}\left(\mathbf{W}_2 \cdot \text{GELU}\left(\mathbf{W}_1 \cdot \text{GAP}\left(\mathbf{F}_u\right)\right)\right) \tag{8}$$

where $\mathbf{W}_1 \in \mathbb{R}^{C\times C/r}$ and $\mathbf{W}_2 \in \mathbb{R}^{C/r\times C}$ are the weights of the fully connected layers, r is the reduction ratio, and sigmoid$(\cdot)$ normalizes the importance scores to the $[0, 1]$ range. The score vector $\mathbf{s} = [s_1, s_2, \ldots, s_C]^\top$, where the ith element s_i quantifies the effective information of the ith band.

The score vector $\mathbf{s}$ provides data-driven prior knowledge: which bands are likely to contain richer information given the current input feature. Instead of applying a simple hard threshold based on these scores, they serve as the core guiding signal for the subsequent adaptive band selection stage.

2) Adaptive Band Selection: Based on the channel importance scores, the band selection state is iteratively optimized through two core operations: a sampling distribution biased toward low-importance bands and a subset-rewriting rule that probabilistically deactivates them, jointly driving efficient convergence to a compact, high-information subset.

a) Sampling distribution construction: In contrast to the fixed proposal distribution employed in classical MCMC, we design a network-guided mixture proposal distribution $q(\mathcal{I}^l|\mathbf{m}^l)$, as detailed below.

On the one hand, based on the current band selection state $\mathbf{m}^l$ (the lth iterative band mask), we perform global exploration by randomly sampling a subset of bands $\mathcal{I}^l \subset \{1, 2, \ldots, C\}$ of fixed size B from a uniform distribution

$$p_{\text{uniform}}\left(\mathcal{I}^l\right) = \frac{1}{\binom{C}{B}} \tag{9}$$

where $\binom{C}{B}$ is the binomial coefficient that equalizes the selection probability over all subsets.

On the other hand, we use the importance score vector $s_i \in \mathbf{s}$ as the guidance weight, assigning higher sampling weights to bands with high importance. Defining the probability of selecting the subset $\mathcal{I}^l$ under the importance guidance as

$$p_{\text{guide}}\left(\mathcal{I}^l\right) = \frac{\prod_{i\in\mathcal{I}^l}(1-s_i)\cdot\prod_{i\notin\mathcal{I}^l}s_i}{\sum_{\mathcal{I}\subset\{1,\ldots,C\},|\mathcal{I}|=B}\prod_{i\in\mathcal{I}}(1-s_i)\cdot\prod_{i\notin\mathcal{I}}s_i}. \tag{10}$$

This distribution prioritizes sampling subsets with more low-importance bands, making them more likely to be rewritten off, which helps to accelerate the convergence of the sampling process.

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We then combine the uniform distribution p_{uniform} with the importance-driven distribution p_{guide} to derive the final adaptive proposal distribution

$$q\left(\mathcal{I}^l \middle| \mathbf{m}^l\right) = \xi \cdot p_{\text{uniform}}\left(\mathcal{I}^l\right) + (1-\xi) \cdot p_{\text{guide}}\left(\mathcal{I}^l\right) \tag{11}$$

where ξ is a balance coefficient, combining random exploration and importance guidance to balance global search and local optimization.

b) Adaptive rewriting mechanism: To prevent the sampling procedure from being trapped in trivial local optima, such as selecting either all bands or none of them, we introduce a rewriting strategy guided by the channel importance scores of the sampled band subset. Under this scheme, bands with higher importance scores are preferentially preserved in the selected set, whereas bands with lower importance scores are more likely to remain excluded.

According to (11), we sample a subset of B bands from the full set of C channels via a mixture adaptive proposal distribution, as follows:

$$\mathcal{I}^l \sim q\left(\mathcal{I}^l \middle| \mathbf{m}^l\right) \quad \forall \mathcal{I}^l \subset \{1, 2, \ldots, C\}, \quad \left|\mathcal{I}^l\right| = B. \tag{12}$$

At each update step, only the components belonging to the subset $\mathcal{I}^l$ are modified, whereas all remaining components are kept identical to their values in the current state $\mathbf{m}^l$.

We introduce the rewriting factor γ_i for deciding whether to rewrite the state of the ith band, which is then constructed by combining the importance score with the state term

$$\gamma_i = \text{sigmoid}\left(\lambda \cdot s_i \cdot \tau\left(m_i^l\right)\right) \tag{13}$$

where $\lambda > 0$ is a preset hyperparameter that controls the strength of the rewrite tendency, and $\tau(m_i^l) = 2m_i^l - 1$ is the state term, which can effectively prevent all bands from being deactivated. The sigmoid function ensures that $\gamma_i \in (0, 1)$, keeping the factor within a valid range.

Since each band is proposed for update independently, we evaluate the rewrite decision separately for every band in the candidate subset $\mathcal{I}^l$. For each band $i \in \mathcal{I}^l$, we sample a random threshold $\epsilon_i \sim \text{Uniform}(0, 1)$ and compare it with γ_i

$$m_i^{l+1} = \begin{cases} 1 - m_i^l, & \text{if } i \in \mathcal{I}^l \text{ and } \epsilon_i \geq \gamma_i \\ m_i^l, & \text{otherwise} \end{cases} \tag{14}$$

and thus forming the updated band selection state

$$\mathbf{m}^{l+1} = \left\{m_i^{l+1}\right\}_{i=1}^{C}. \tag{15}$$

The resulting state is propagated to the subsequent batch-sampling iteration, thereby achieving state updates that maintain computational efficiency through mini-batch operations while minimally affecting the condition of full-band coverage.

3) Iterative Sampling and Feature Refinement: The entire sampling process is an iterative convergence process with L iterations. Ergodic coverage via iterative batch sampling ensures that all input bands are traversed during inference; randomness only affects the order or local combinations, preventing persistent omission of channels. In addition, adaptive, prior-guided band selection provides mild input-dependent guidance while preserving exploration. Each iteration consists of three steps: proposal subset generation, rewriting factor evaluation, and band state update.

a) Iterative Sampling Process: The band selection state $\mathbf{m}^1$ is initialized as an all-ones vector. For each batch sampling iteration $l = 1, 2, \ldots, L$, the specific steps are as follows.

Step 1: Proposal Subset Generation: Based on the current state $\mathbf{m}^l$ and the importance score vector $\mathbf{s}$ output by the network $\Phi(\cdot)$, we generate the proposal subset $\mathcal{I}^l$ through the adaptive proposal distribution $q(\mathcal{I}^l|\mathbf{m}^l)$.

Step 2: Rewriting Factor Evaluation: For each band in the sampled subset $\mathcal{I}^l$, we combine its current state m_i^l with the importance score s_i to compute the corresponding band-state rewrite factor γ_i using (13).

Step 3: Band State Update: We sample a random threshold ϵ_i for each rewrite factor γ_i and update the band subset according to (14). This yields the band selection state $\mathbf{m}^{l+1}$ for the lth batch sampling process.

After L iterations, the sampling process converges to the optimal band selection state $\mathbf{m}^*$. We then use this state vector as the band mask to perform dimensionality reduction on the upsampled feature $\mathbf{F}_u$. The dimensionality reduction process is realized by element-wise multiplication and a convolution operation to adjust the feature dimension

$$\mathbf{F}_d = \text{Conv}\left(\mathbf{F}_u \odot \mathbf{m}^*\right) \in \mathbb{R}^{H \times W \times c} \tag{16}$$

where $c \ll C$ is the number of bands after dimensionality reduction, and $\text{Conv}(\cdot)$ is a 1×1 convolutional layer, which is used to map the feature dimension from C to c.

b) Reduced feature refinement: To efficiently model long-range dependencies within the compressed feature $\mathbf{F}_d$, following the SGN strategy in [59] (Section II-C), we introduce a sorted state-space model (Sort-SS2D). As shown in Fig. 3(c), the process first projects features into a semantic space and assigns a discrete semantic category to each pixel using the Softmax-Gumbel function. All pixels are then reordered according to their token categories to construct a semantically coherent sequence. This ordered sequence is fed into a lightweight SS2D module for unidirectional scanning to capture long-range dependencies. After scanning, the sequence is restored to its original spatial order, resulting in a globally context-enhanced feature $\mathbf{F}_{spe}$. The formula is described as follows:

$$\mathbf{F}_{spe} = \text{Sort-SS2D}\left(\text{LN}\left(\mathbf{F}_d\right)\right). \tag{17}$$

Finally, to compensate for information loss during dimensionality reduction, the feature $\mathbf{F}_d$ is concatenated with $\mathbf{F}_{spe}$ along the channel dimension. Then, fuses them while restoring the channel dimension to the original size. This feature is further refined by multilayer perceptron (MLP)

$$\mathbf{X}^t = \text{MLP}\left(\text{LN}\left(\text{Concat}\left(\mathbf{F}_d, \mathbf{F}_{spe}\right)\right)\right) + \mathbf{F}_u \tag{18}$$

where $\mathbf{X}^t$ serves as the final output of the S2C and is fed into the subsequent S2M module.

D. S2M Module

To achieve two-stage lightweighting in synergy with the S2C module, we propose an S2M module for adaptive spatial dimensionality reduction. As illustrated in Fig. 3(b) and

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Algorithm 2 Overall Framework of S2M

Input: $\mathbf{X}^t \in \mathbb{R}^{H\times W\times C}, \mathbf{m}^* \in \mathbb{R}^C$
% Band State Guide
$\mathbf{F}_s = \mathbf{X}^t \odot \mathbf{m}^*$
% Semantic-aware Token Merging
partitioning $\mathbf{F}_s$ into windowed feature $\mathbf{F}_w$ via (20)
get each window centroids $\mathbf{F}_p$ via (21)
modeling intrawindow semantic similarity via (23)
get learnable assignment matrix $\mathbf{M}$ via (24)
assembling spatial reduction map $\mathbf{M}^t_{spa}$ with (25)
merging similar tokens $\mathbf{F}_m$ via (26)
% Merging Feature Refinement
$\mathbf{F}_{spa} = \text{Sort-SS2D}(\text{LN}(\mathbf{F}_m))$
$\mathbf{F}^{t+1} = \text{MLP}(\text{LN}(\text{Concat}(\mathbf{F}_m, \mathbf{F}_{spa}))) + \mathbf{F}_m$
Output: $\mathbf{F}^{t+1}$

Algorithm 2, the S2M operates in three stages: 1) utilizing the band selection state from S2C to focus the input features on informative bands, ensuring cross-dimensional synergy; 2) merging perceptually similar pixels within local windows based on spectral affinity to reduce spatial redundancy; and 3) refining the merged tokens by instilling global contextual semantics via the Sort-SS2D block.

1) Band State Guide: The S2M module first receives and applies the band selection state $\mathbf{m}^*$ from the S2C. Similar to (16), the $\mathbf{m}^*$ is multiplied channel-wise with the current input feature $\mathbf{X}^t$ to yield a spectrally selected feature

$$\mathbf{F}_s = \mathbf{X}^t \odot \mathbf{m}^*. \tag{19}$$

This step ensures that subsequent spatial processing concentrates on the important bands already identified by the band selection mechanism.

2) Semantic-Aware Token Merging: The S2M module then establishes a similarity distribution between each centroid and all pixel tokens, from which a dynamic mapping matrix is derived to merge the pixel tokens.

a) Subwindow centroid construction: To reduce the computational complexity of subsequent operations, the input feature $\mathbf{F}_s \in \mathbb{R}^{H\times W\times C}$ is first partitioned into N_w nonoverlapping local windows, each of size $n \times n$. This yields the windowed feature

$$\mathbf{F}_w = \text{Partition}(\mathbf{F}_s) \in \mathbb{R}^{N_w\times K\times C} \tag{20}$$

where $K = n^2$ is the number of pixels per window, and $N_w = HW/K$ denotes the total number of windows. To initialize merging centroids, a pooling operation with stride s is applied within each window, generating k initial centroids per window

$$\mathbf{F}_p = \text{Pooling}(\mathbf{F}_w) \in \mathbb{R}^{N_w\times k\times C} \tag{21}$$

where $k = \lfloor n/s \rfloor^2$ is the number of centroids per window. We then focus on the operations within a single window. The pixel tokens and initial centroids for an arbitrary window are denoted as $\mathbf{W}^{(n_w)} \in \mathbb{R}^{K\times C}$ and $\mathbf{C}^{(n_w)} \in \mathbb{R}^{k\times C}$, which are slices of $\mathbf{F}_w$ and $\mathbf{F}_p$, respectively, where n_w denotes the window index.

b) Similarity mapping modeling: To quantify the semantic relationship between pixels $\mathbf{F}_w$ and centroids $\mathbf{F}_p$, we model their spectral similarity distribution within each window. For each pixel token $w_i \in \mathbb{R}^{1\times 1\times C}$ (the ith row of $\mathbf{W}^{(n_w)}$) and each pooling centroid $c_j \in \mathbb{R}^{1\times 1\times C}$ (the jth row of $\mathbf{C}^{(n_w)}$), we compute their spectral similarity using the spectral angle mapper (SAM), a metric well-suited for hyperspectral feature comparison. The resulting similarity matrix $\mathbf{S}^{(n_w)} \in \mathbb{R}^{K\times k}$ is defined element-wise as

$$S_{i,j} = \text{SAM}(w_i, c_j) = 1 - \frac{1}{\pi}\arccos\left(\frac{w_i \cdot c_j}{\|w_i\| \cdot \|c_j\|}\right) \tag{22}$$

where $S_{i,j}$ denotes the element in the ith row and jth column of the matrix $\mathbf{S}^{(n_w)}$; a value of 1 indicates completely identical spectral features.

This similarity matrix is then normalized for each centroid c_j to convert the similarity scores $S_{i,j}$ into a probability distribution that captures pixel-to-centroid assignments. The probability distribution matrix $\mathbf{P}^{(n_w)} \in \mathbb{R}^{K\times k}$ is obtained as

$$P_{i,j} = \frac{S_{i,j}}{\sum_{i=1}^{K} S_{i,j}} \in \mathbf{P}^{(n_w)}. \tag{23}$$

Similarly, $P_{i,j}$ denotes the element in the ith row and jth column of the matrix $\mathbf{P}^{(n_w)}$. Each element $P_{i,j}$ represents the probability that pixel w_i is assigned to centroid c_j, thereby forming a structured assignment prior to that serves as the foundation for the subsequent learnable merging process.

c) Similarity-driven token merging: To adaptively adjust the merging process according to input content, we transform the similarity probability distribution $\mathbf{P}^{(n_w)}$ into a learnable assignment matrix. This is achieved by introducing two learnable parameters, enabling dynamic sensitivity adjustment. The mapping matrix $\mathbf{M}^{(n_w)}$ is computed as

$$\mathbf{M}^{(n_w)} = \text{sigmoid}\left(\alpha \cdot \mathbf{P}^{(n_w)} + \beta\right) \tag{24}$$

where sigmoid$(\cdot)$ ensures each element of $\mathbf{M}^{(n_w)}$ lies in the range $(0, 1)$. This matrix represents the assignment weight from each pixel to each candidate centroid, with α controlling the sharpness of the assignment and β providing a bias term.

The mapping matrices from all windows are assembled into a global spatial reduction map $\mathbf{M}^t_{spa} \in \mathbb{R}^{HW\times hw}$. This is formally expressed as constructing a block diagonal matrix

$$\mathbf{M}^t_{\text{spa}} = \begin{bmatrix} \mathbf{M}^{(1)} & & \\ & \ddots & \\ & & \mathbf{M}^{(N_w)} \end{bmatrix} \tag{25}$$

where t denotes the number of module iterations, $HW = N_w \cdot K$ and $hw = N_w \cdot k$, and each diagonal block $\mathbf{M}^{(n_w)} \in \mathbb{R}^{K\times k}$ is the mapping matrix of the ith window, and off-diagonal blocks are zero matrices. This structure ensures that each pixel only interacts with centroids within its own window, preserving local computation while enabling global representation of the assignment mapping.

Finally, this block diagonal matrix $\mathbf{M}^t_{\text{spa}}$ serves as an assignment guide for merging pixels across the entire feature map. Inspired by the value projection in self-attention, we define a transformation $\mathcal{T}(\cdot)$ that consists of a convolutional layer followed by reshaping to appropriate dimensions. Specifically,

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$\mathcal{T}(\mathbf{F}_w) \in \mathbb{R}^{HW\times C}$ and $\mathcal{T}(\mathbf{F}_p) \in \mathbb{R}^{hw\times C}$. The merged centroids are then obtained as

$$\mathbf{F}_m^{\top} = \frac{1}{N_m}\left(\mathcal{T}(\mathbf{F}_w)^{\top} \times \mathbf{M}_{spa}^{t} + \mathcal{T}(\mathbf{F}_p)^{\top}\right) \tag{26}$$

$$N_m = \left(1 + \text{sum}\left(\mathbf{M}_{spa}^{t}\right)\right) \tag{27}$$

where N_m is the normalization factor, keeping the initial centroid and stabilizing aggregation, and sum$(\cdot)$ denotes the sum of all elements of the matrix.

3) Merging Feature Refinement: To model long-range spatial dependencies and infuse the merged features with global semantic context, $\mathbf{F}_m$ is fed into the Sort-SS2D for processing. As described in Section III-C3, it efficiently constructs and processes a globally semantically correlated sequence through a series of operations: semantic token assignment, sequence reordering, unidirectional scanning via SS2D, and sequence restoration. Its output is the feature $\mathbf{F}_{spa}$, which is enhanced with both spatial and semantic contexts

$$\mathbf{F}_{spa} = \text{Sort-SS2D}\left(\text{LN}\left(\mathbf{F}_m\right)\right). \tag{28}$$

Finally, the enhanced feature $\mathbf{F}_{spa}$ is concatenated with $\mathbf{F}_m$, then passed through LayerNorm and an MLP for nonlinear transformation and refinement, outputting the final result of the S2M

$$\mathbf{F}^{t+1} = \text{MLP}\left(\text{LN}\left(\text{Concat}\left(\mathbf{F}_m, \mathbf{F}_{spa}\right)\right)\right) + \mathbf{F}_m \tag{29}$$

where $\mathbf{F}^{t+1} \in \mathbb{R}^{h\times w\times C}$ and is fed into S2C module for the next iteration. By sharing the spectral selection prior with S2C and executing a progressive spatial lightweighting from local merging to global semantic fusion, the S2M block jointly forms an efficient and collaborative two-stage processing framework.

IV. Experiments

A. Datasets and Evaluation Metrics

To validate the generalization and effectiveness of the proposed method across different scenarios, we perform experiments on four categories of scenes: natural scenes using the CAVE [62] dataset,[1] remote sensing scenes with the Pavia Center[2] and Chikusei [63] datasets,[3] real-world scenes with the Foster [64] dataset,[4] and a dedicated noise robustness experiment on specific bands.

1) CAVE: The CAVE dataset was acquired with a cooled CCD camera and a tunable optical filter, containing 32 indoor hyperspectral scenes. Each scene has a 512 × 512 pixel resolution and a spectral range of 400–700 nm, with 31 bands at 10 nm intervals.

2) Chikusei: The Chikusei dataset was acquired with the Headwall Hyperspec-VNIR-C sensor, covering about 5 km^2 of Chikusei, Japan. It has a spatial resolution of 2.5 m and a size of 2517 by 2335 pixels, with a spectral range from 363 to 1018 nm and 128 bands at 5 nm intervals.

[1]https://www.cs.columbia.edu/CAVE/databases/multispectral/

[2]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

[3]https://www.sal.t.u-tokyo.ac.jp/hyperdata

[4]https://personalpages.manchester.ac.uk/staff/d.h.foster/Hyperspectral_images_of_natural_scenes_02.html

3) Pavia Center: The Pavia Center dataset was captured by the ROSIS-03 sensor over the city center of Pavia, northern Italy, covering approximately 1.3 km^2. With a spatial resolution of 1.3 m and an original image size of 1096 × 1096 pixels, the dataset covers a spectral range from 430 to 860 nm encompassing 102 valid bands.

4) Foster: The Foster dataset was collected by the University of Minho in Portugal during the Spring and Summer of 2002–2003. It comprises 30 natural outdoor scenes with a spatial resolution of 1204 × 1344 pixels and a spectral range of 400–720 nm, sampled every 10 nm into 33 contiguous narrow bands. All images were acquired using a low-noise, Peltier-cooled Hamamatsu C4742-95-12ER camera.

5) Evaluation Metrics: We conduct a comprehensive quantitative evaluation of all experimental results using six widely adopted metrics: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [65], SAM [66], erreur relative globale adimensionnelle de synthese (ERGAS) [67], root-mean-squared error (RMSE), and CC [68]. Their optimal values are $+\infty$, 1, 0, 0, 0, and 1, respectively.

B. Comparison Methods

In this study, six deep learning-based HSI-SR methods are selected for comparative analysis: two classical methods (MCNet [10] and SSPSR [11]), two state-of-the-art methods (MambaHSISR [29] and VolFormer [24]), and two lightweight methods (ESSAFormer [54] and CATANet [69]). Traditional bicubic interpolation is also included as a nonlearning baseline. Specifically, MCNet and SSPSR are seminal works that introduced CNN architectures and residual learning to HSI-SR and have been widely adopted as benchmarks. MambaHSISR and VolFormer represent the latest advances specifically tailored for HSI-SR tasks, reflecting the current research frontier. ESSAFormer and CATANet, similar to our proposed method, emphasize the balance between efficiency and performance, enabling direct comparison of both accuracy and computational cost; notably, CATANet is a lightweight method directly transferred from RGB-SR to HSI-SR. To ensure a fair comparison, all methods were reimplemented and trained on identical datasets and degradation conditions, following the original hyperparameter configurations and optimization strategies.

C. Implementation Details

Data preparation follows extensive settings [11], [24], [54]. During the training phase, we extract overlapping patches from the training images for data augmentation. For the ×2 and ×4 super-resolution tasks, the patch size is set to 64 × 64 pixels; for the ×3 task, it is set to 63 × 63 pixels. The stride for all settings is 32 pixels. For the CAVE dataset, we randomly select 12 images for testing and use the remaining 20 for training. For the Chikusei dataset, following common practice, we first center-crop the original image to a 2048 × 2048 region, which is then divided into 16 nonoverlapping 512 × 512 subimages. The top four subimages are selected as the test set, with the remaining 12 used for training. For the Pavia center dataset, considering its original image is stitched from left and right parts, we select a contiguous area on the left side, divided

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TABLE I
HYPERPARAMETER TUNING RESULTS OF ITERATIONS T FOR $\times 4$

Datasets	$T=1$	$T=2$	$T=3$	$T=4$	$T=5$	$T=8$	$T=10$
Chikusei	39.0234	39.5127	39.6519	**39.6645**	39.6602	39.6581	39.6505
Pavia	28.8156	29.1589	**29.3146**	29.3012	29.2987	29.2801	29.2775
CAVE	36.2843	36.5234	36.6905	**36.7023**	36.7001	36.6820	36.6752
Average PSNR (dB)	34.7078	35.0650	35.2190	**35.2227**	35.2197	35.2067	35.2011
Performance Gain	-	**+0.3572**	**+0.1540**	**+0.0037**	-0.0030	-0.0130	-0.0056

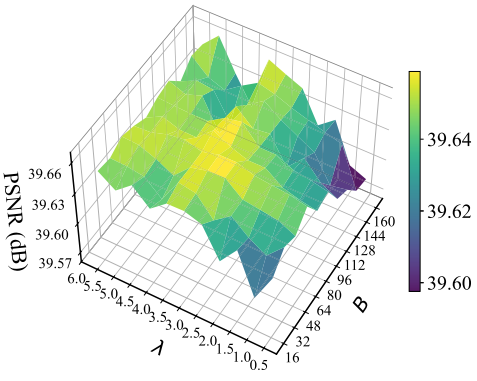


Fig. 4. Hyperparameters tuning results of λ and B.

into nonoverlapping 216×216 blocks as the test set, while the larger right-side area is used for training. To simulate a realistic imaging degradation process, the LR images are generated through the following steps. First, the HR image is blurred by applying an anisotropic Gaussian kernel. The kernel size is 19×19, with the kernel width randomly sampled from the range $[0.4, 5]$ and the rotation angle randomly sampled from $[0, \pi]$. Subsequently, the blurred image undergoes bicubic downsampling by the specified factor to produce the final LR image.

We train the proposed S^{2}CD-Net end-to-end using the PyTorch framework and the Adam optimizer. The L1 loss serves as the loss function. The initial learning rate is set to 2×10^{-4} and is decayed to 1×10^{-5} using a cosine annealing scheduler. The shallow feature extraction layer projects the input into a channel dimension of $C = 256$. Random rotation and flipping are applied during training for better generalization. Shown as Table I, PSNR rises sharply from $T = 1$ to 2, then saturates beyond $T = 3$, considering performance and efficiency, we set $T = 3$. In the S2C module, the mixture distribution balance coefficient is $\xi = 0.3$, the batch sampling iteration count is $L = 5$, the number of bands sampled per iteration is $B = 64$, and the state rewriting strength parameter is $\lambda = 3.5$ (the tuning analysis for λ and B is shown in Fig. 4). In the S2M module, the window size for spatial merging is set to $n = 2 \times scale$, and the merging ratio is $s = scale$, where $scale$ is the super-resolution factor; the learnable parameters are initialized as $\alpha = 1$ and $\beta = 0$. All experiments are conducted on NVIDIA GeForce RTX 4090 GPU.

D. Experimental Results

In this part, the performance of the proposed method is comprehensively evaluated from both quantitative metrics and qualitative visual quality. The best quantitative results are highlighted in bold, while the suboptimal ones are indicated with an underline.

1) Pavia Center: The Pavia center dataset contains complex urban scenes. As shown in Table II (left side), our S^{2}CD-Net demonstrates strong competitiveness. For the $\times 2$ task, it achieves optimal PSNR and SSIM values of 34.24 dB and 0.9393, respectively. Simultaneously, its computational cost (14.19G FLOPs) is less than 10% of that of the suboptimal nonlightweight SOTA method, achieving leading performance in both accuracy and efficiency. As the scale factor increases further, the advantages of our method in terms of structural similarity and error suppression become more pronounced. This indicates that our method can effectively handle complex edges, textures, and mixed pixels in urban scenes without sacrificing reconstruction accuracy while significantly reducing the computational burden. The visual comparison results are shown in Fig. 5. The latest transformer-based methods restore some structures but still show shortcomings in reconstructing building edges. In contrast, our method better restores sharp building edges and roof details. Furthermore, as shown in the error maps in the third row, the error magnitudes in high-frequency regions are significantly lower, particularly in high-frequency detail areas, demonstrating its excellent spatial detail reconstruction capability.

2) Chikusei: As shown in Table II (right side), our method achieves the best performance on the majority of metrics across all three scaling factors. For the $\times 2$ task, it leads across all six metrics compared to the current SOTA method, VolFormer. It maintains stable advantages in the $\times 3$ and $\times 4$ tasks as well. Notably, while achieving high performance, the FLOPs and parameter count of S^{2}CD-Net are significantly lower than those of nonlightweight SOTA methods. Fig. 6 presents the visual comparison results. Compared to the blurred results from bicubic and CNN-based methods and the noticeable distortion in CATANet due to its direct adaptation from the RGB to the HSI domain, our method reconstructs continuous road textures and clearer architectural details. The error maps show that our method yields smaller and more uniformly distributed reconstruction errors, especially in building areas. This directly validates the effectiveness of the proposed spectral–spatial-coupled dimensionality reduction mechanism: through dynamic spectral screening and semantically aware spatial merging, it drastically reduces data redundancy with low computational overhead while accurately preserving the information crucial for reconstruction, achieving high-fidelity reconstruction with high efficiency.

3) CAVE: The CAVE dataset contains natural scene objects with smooth spectral curves and rich spatial textures. As shown in Table III, our method leads comprehensively at lower scaling factors. At higher scaling factors, it maintains leading or comparable results in PSNR, SSIM, and CC, while often performing better in SAM, ERGAS, and RMSE. S^{2}CD-Net maintains performance comparable to nonlightweight methods despite requiring extremely low computational resources. This reflects the design motivation of the method: the front-end adaptive processing effectively reduces the complexity and burden of subsequent global modeling, enabling even lightweight backend modeling units to obtain high-quality

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TABLE II

QUANTITATIVE COMPARISON RESULTS OF DIFFERENT HSI-SR METHODS ON THE PAVIA AND CHIKUSEI REMOTE SENSING DATASETS

Method	Scale	Pavia								Chikusei							
		FLOPs	Params	PSNR↑	SSIM↑	SAM↓	ERGAS↓	RMSE↓	CC↑	FLOPs	Params	PSNR↑	SSIM↑	SAM↓	ERGAS↓	RMSE↓	CC↑
Bicubic	×2	N/A	N/A	28.3000	0.7309	5.8933	10.2475	0.0414	0.8736	N/A	N/A	38.4076	0.8997	3.2809	5.3185	0.0149	0.9235
SSPSR	×2	134.02G	18.95M	33.4465	0.9215	4.4133	5.6421	0.0234	0.9520	166.94G	19.13M	43.6421	0.9770	1.9463	2.7474	0.0076	0.9791
MCNet	×2	262.73G	1.93M	33.0845	0.9182	4.5210	5.9437	0.0241	0.9487	329.70G	1.93M	42.8715	0.9759	2.1054	2.8871	0.0080	0.9772
ESSAFormer	×2	52.89G	9.01M	33.8650	0.9290	4.2969	5.3258	0.0217	0.9596	53.20G	9.13M	44.2404	0.9782	1.8330	2.6381	0.0071	0.9800
MambaHSISR	×2	104.35G	102.54M	34.0241	0.9315	4.2989	5.2041	0.0227	0.9546	104.56G	102.65M	44.2688	0.9770	1.8297	2.6244	0.0073	0.9814
VolFormer	×2	389.01G	29.64M	34.1725	0.9359	4.0026	5.1181	0.0210	0.9622	488.41G	29.82M	44.4210	0.9778	1.8002	2.5862	0.0071	0.9819
CATANet	×2	15.34G	13.29M	34.1521	0.9341	4.1356	5.2875	0.0225	0.9558	15.65G	13.41M	44.3259	0.9786	1.9109	2.5937	0.0071	0.9805
Ours	×2	14.19G	4.18M	**34.2413**	**0.9393**	**3.9014**	**5.0241**	**0.0197**	**0.9667**	14.49G	4.30M	**44.6211**	**0.9803**	**1.7679**	**2.4807**	**0.0068**	**0.9822**
Bicubic	×3	N/A	N/A	27.3650	0.6890	6.2645	11.1808	0.0452	0.8518	N/A	N/A	37.2710	0.8834	3.6333	5.8728	0.0165	0.9098
SSPSR	×3	60.06G	21.90M	30.5127	0.8711	5.2431	7.5681	0.0308	0.9242	74.50G	22.08M	40.8804	0.9552	2.4722	3.9784	0.0108	0.9587
MCNet	×3	202.53G	2.04M	30.4852	0.8695	5.3240	7.6127	0.0321	0.9187	254.15G	2.04M	40.5951	0.9523	2.5889	4.0012	0.0116	0.9510
ESSAFormer	×3	48.99G	12.34M	31.1448	0.8802	4.9799	7.1519	0.0285	0.9357	49.26G	12.46M	41.3715	0.9577	2.3301	3.6292	0.0100	0.9644
MambaHSISR	×3	45.52G	103.28M	31.1580	0.8822	4.9237	7.2412	0.0283	0.9324	45.68G	103.39M	41.5681	0.9584	2.3252	3.6021	0.0100	0.9657
VolFormer	×3	179.02G	29.62M	31.2452	0.8845	4.8422	7.0168	0.0280	**0.9390**	222.56G	29.80M	41.7952	0.9614	2.1982	3.4912	0.0097	0.9671
CATANet	×3	8.04G	16.24M	29.8716	0.8416	5.7539	8.2941	0.0333	0.9147	8.31G	16.36M	39.1185	0.9312	3.7701	4.7681	0.0128	0.9437
Ours	×3	12.97G	7.40M	**31.3790**	**0.8862**	**4.6355**	**6.9703**	**0.0277**	0.9388	13.24G	7.52M	**41.9310**	**0.9626**	**2.1354**	**3.3810**	**0.0094**	**0.9686**
Bicubic	×4	N/A	N/A	26.6149	0.6351	6.6156	12.1444	0.0493	0.8231	N/A	N/A	36.4049	0.8630	3.9939	6.4297	0.0180	0.8939
SSPSR	×4	52.26G	21.31M	28.0846	0.7864	5.7764	9.5167	0.0387	0.8874	63.37G	21.49M	38.8095	0.9297	2.8641	4.9781	0.0135	0.9384
MCNet	×4	230.51G	2.17M	28.1734	0.7822	5.7720	9.4510	0.0374	0.8894	289.26G	2.17M	38.3246	0.9165	3.1529	5.4868	0.0147	0.9259
ESSAFormer	×4	50.19G	11.52M	29.1755	0.8077	5.5532	9.0062	0.0360	0.8993	50.45G	11.64M	39.5183	0.9334	2.8268	4.4465	0.0126	0.9469
MambaHSISR	×4	27.05G	103.13M	29.1845	0.8087	5.5516	9.0008	0.0359	0.8952	27.20G	103.24M	39.5871	0.9338	2.7781	4.7821	0.0124	0.9448
VolFormer	×4	113.98G	29.79M	29.2386	0.8108	5.4592	**8.7826**	0.0357	0.9007	140.04G	29.97M	39.6264	0.9355	2.7524	**4.3761**	0.0121	**0.9497**
CATANet	×4	6.44G	15.65M	28.0369	0.7539	6.0629	10.2823	0.0413	0.8694	6.70G	15.77M	38.1116	0.9097	3.1852	5.2581	0.0146	0.9284
Ours	×4	14.29G	6.54M	**29.3146**	**0.8127**	**5.3402**	8.8377	**0.0354**	**0.9028**	14.55G	6.66M	**39.6519**	**0.9361**	**2.6268**	4.4106	**0.0119**	0.9488

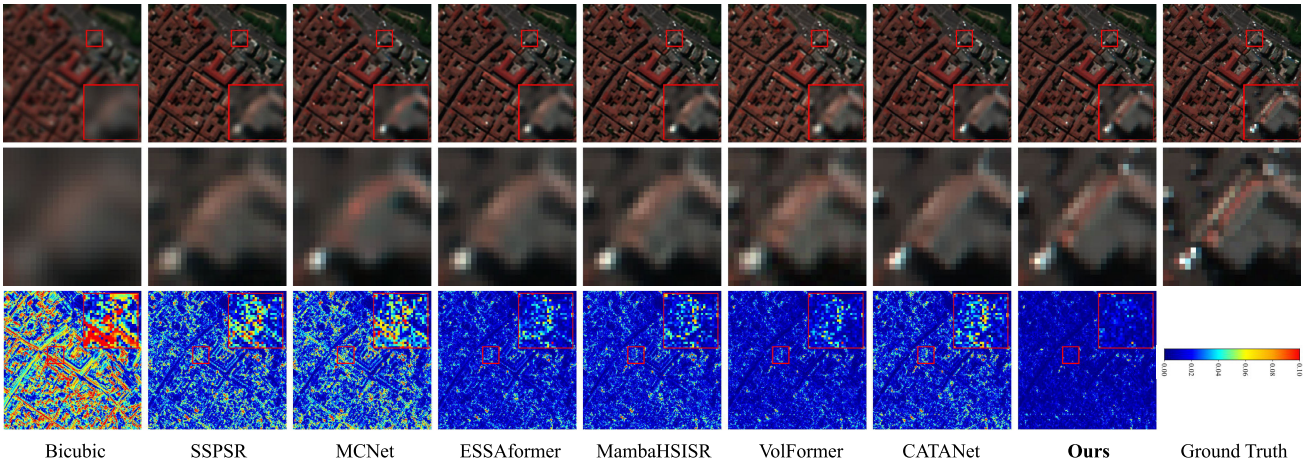


Fig. 5. Visual comparison of the perceptual quality of all competing methods on the Pavia dataset for ×2 upscaling. Super-resolution results (first row), detailed patches (second row), and error maps (third row).

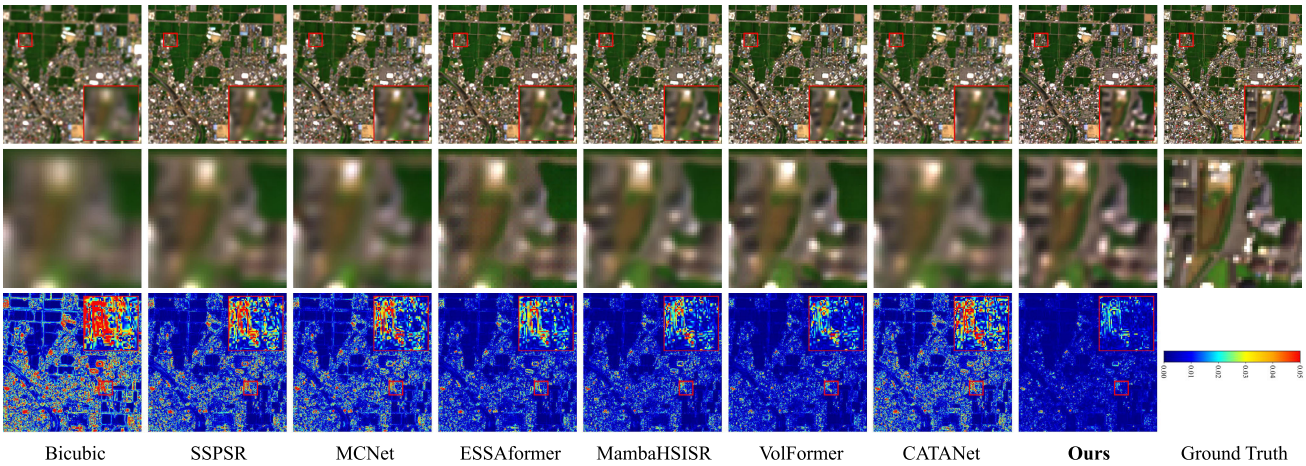


Fig. 6. Visual comparison of the perceptual quality of all competing methods on the Chikusei dataset for ×4 upscaling. Super-resolution results (first row), detailed patches (second row), and error maps (third row).

information representation, ultimately achieving a new balance between performance and efficiency. As shown in Fig. 7, our method's results are closest to the ground truth image in terms of texture clarity, edge sharpness, and color continuity, with

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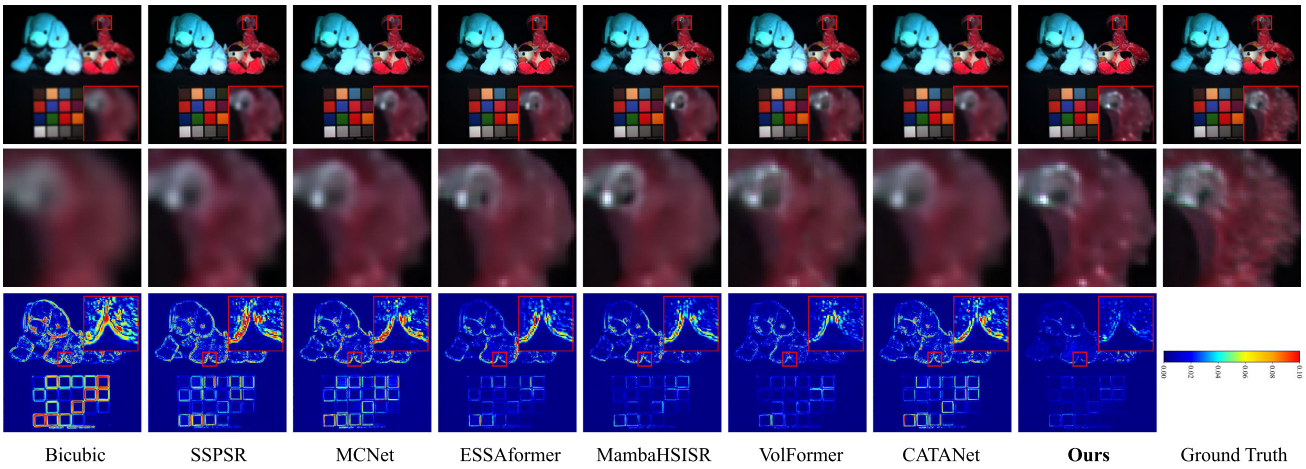


Fig. 7. Visual comparison of the perceptual quality of all competing methods on the CAVE dataset for ×3 upscaling. Super-resolution results (first row), detailed patches (second row), and error maps (third row).

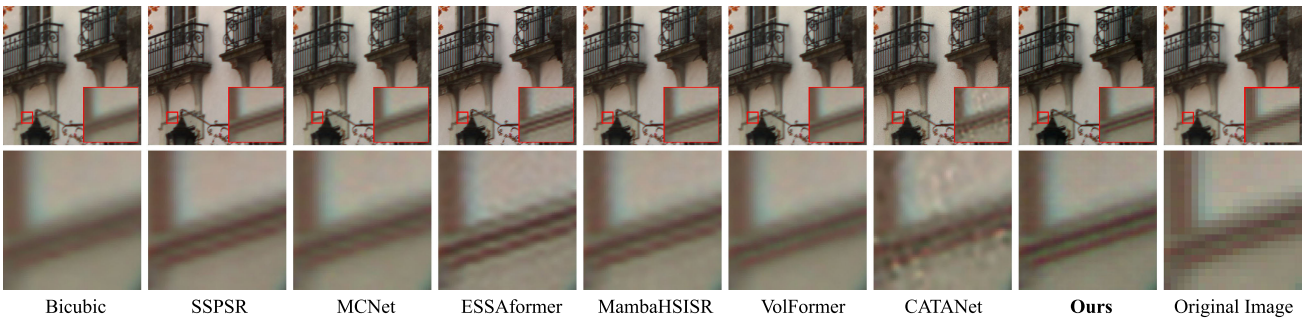


Fig. 8. Visual comparison of the perceptual quality of all competing methods on the real-world dataset for ×4 upscaling (top) super-resolution results and (bottom) detailed patches.

TABLE III

QUANTITATIVE COMPARISON RESULTS OF DIFFERENT HSI-SR METHODS ON THE CAVE NATURAL DATASETS

Method	Scale	CAVE							
		FLOPs	Params	PSNR↑	SSIM↑	SAM↓	ERGAS↓	RMSE↓	CC↑
Bicubic	×2	N/A	N/A	36.4903	0.9230	3.9005	6.1201	0.0183	0.9777
SSPSR	×2	51.59G	18.45M	39.1484	0.9551	3.7117	4.4713	0.0138	0.9864
MCNet	×2	79.85G	1.93M	39.7542	0.9564	3.6802	4.2124	0.0133	0.9870
ESSAFormer	×2	52.05G	8.69M	39.9224	0.9576	3.6705	4.0096	0.0121	0.9874
MambaHSISR	×2	103.78G	102.23M	40.5573	0.9580	3.5441	4.0012	0.0118	0.9888
VolFormer	×2	140.32G	29.15M	40.6020	0.9608	3.5254	3.8677	0.0118	0.9897
CATANet	×2	14.50G	12.97M	40.2150	0.9578	3.6909	4.0003	0.0121	0.9890
Ours	×2	13.35G	3.86M	**40.7917**	**0.9629**	**3.4847**	**3.6829**	**0.0112**	**0.9912**
Bicubic	×3	N/A	N/A	35.3680	0.9137	4.2810	6.6894	0.0201	0.9753
SSPSR	×3	23.83G	21.41M	37.1455	0.9430	4.1347	5.1241	0.0157	0.9850
MCNet	×3	61.55G	2.04M	37.5248	0.9455	3.8757	4.9952	0.0152	0.9857
ESSAFormer	×3	48.25G	12.01M	37.8051	0.9458	3.8752	4.8258	0.0147	0.9870
MambaHSISR	×3	45.10G	102.97M	38.4210	0.9452	3.7062	4.7981	0.0144	0.9857
VolFormer	×3	70.00G	29.13M	38.6581	0.9467	**3.4542**	**4.4022**	0.0140	0.9873
CATANet	×3	7.32G	15.92M	37.5997	0.9448	4.0599	4.9351	0.0149	0.9870
Ours	×3	12.23G	7.07M	**38.7483**	**0.9496**	3.4863	4.4438	**0.0137**	**0.9879**
Bicubic	×4	N/A	N/A	33.9353	0.9026	4.5091	7.5779	0.0226	0.9720
SSPSR	×4	24.34G	20.82M	35.9524	0.9278	4.4241	6.1124	0.0184	0.9781
MCNet	×4	70.06G	2.17M	36.1698	0.9314	4.3257	5.9254	0.0176	0.9789
ESSAFormer	×4	49.48G	11.19M	36.3695	0.9310	4.3550	5.6636	0.0171	0.9815
MambaHSISR	×4	26.66G	102.82M	36.4810	0.9311	4.3591	5.6620	0.0168	0.9799
VolFormer	×4	48.65G	29.30M	36.6603	0.9323	**4.1564**	5.6263	**0.0163**	0.9830
CATANet	×4	5.73G	15.33M	36.4490	0.9271	4.4874	5.6860	0.0171	0.9825
Ours	×4	13.58G	6.22M	**36.6905**	**0.9337**	4.3197	**5.5442**	0.0168	**0.9835**

TABLE IV

RECONSTRUCTION PERFORMANCE OF INJECTING DIFFERENT NOISE LEVELS INTO DIFFERENT IMPORTANCE BAND GROUPS

Band Group	Noise Level	×2			×4		
		Chikusei	Pavia	CAVE	Chikusei	Pavia	CAVE
Low-importance	0.05	44.6215	34.2418	40.7921	39.6530	29.3144	36.6908
	0.10	44.6231	34.2432	40.7932	39.6542	29.3147	36.6918
	0.15	44.6233	34.2438	40.7940	39.6546	29.3151	36.6919
Medium-importance	0.05	44.6132	34.2387	40.7842	39.6462	29.3102	36.6841
	0.10	44.6065	34.2341	40.7768	39.6408	29.3061	36.6772
	0.15	44.5902	34.2272	40.7621	39.6325	29.2997	36.6661
High-importance	0.05	44.6056	34.2258	40.7785	39.6311	29.2994	36.6703
	0.10	44.5876	34.2135	40.7689	39.6283	29.2868	36.6612
	0.15	44.5658	34.1946	40.7475	39.6156	29.2785	36.6523

the fine texture of the toy's fur being delicately reproduced. Errors are minimal in high-frequency texture and edge regions, aligning with the characteristics of human visual perception.

4) Real-World: Fig. 8 presents the visual comparison results in real-world scenarios. Under unknown and complex degradation conditions, methods based on fixed architectures or strong data assumptions generally exhibit residual blurring or signs of artifacts in the edge areas. In contrast, the reconstruction results of S^{2}CD-Net demonstrate superior visual clarity, without noticeable or mismatched artifacts. This advantage directly stems from the stronger generalization ability of our method's adaptive mechanisms when facing unknown degradations.

5) Robustness to Noise in Specific Bands: We divide bands into high-, medium-, and low-information groups based on learned importance and inject Gaussian noise with $\sigma \in \{0.05, 0.10, 0.15\}$ into each group. As shown in Table IV, noise in low-importance bands has a negligible effect and even slightly encourages the model to rely more on informative bands. Noise in medium- and high-importance bands gradually degrades performance, but severely corrupted high-importance bands are effectively suppressed by the network, limiting performance loss and demonstrating robustness to noise perturbations.

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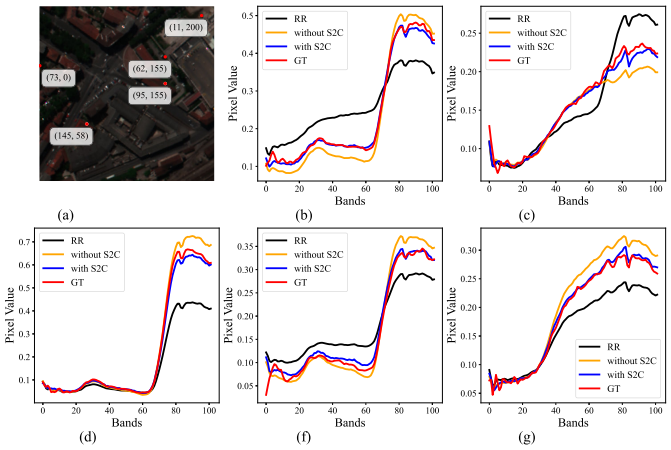


Fig. 9. Ablation study of S2C on the Pavia dataset presents spectral curves from five randomly chosen points. (a) HR HSI with points. (b) Point (11, 200). (c) Point (95, 155). (d) Point (73, 0). (f) Point (62, 155). (g) Point (145, 58).

TABLE V
ABLATION STUDY ON THE EFFECTIVENESS OF S2C MODULE

Dataset	Component	FLOPs	Params	PSNR↑	SAM↓
CAVE	Random Rewriting	13.58G	6.22M	35.4431	4.9681
	w/o. S2C	47.96G	8.98M	36.5812	4.3614
	w. S2C	13.58G	6.22M	36.6905	4.3197
Chikusei	Random Rewriting	14.55G	6.66M	38.0528	3.0241
	w/o. S2C	48.93G	9.43M	39.4972	2.6887
	w. S2C	14.55G	6.66M	39.6519	2.6268

E. Ablation Study

1) Ablation of S2C: As shown in Table V, we compared three configurations: 1) random rewriting, which replaces the adaptive rewriting mechanism with random band selection and fails to differentiate band information, resulting in the worst performance; 2) w/o. S2C, where processing in the original full-band domain leads to a sharp increase in computational cost while still delivering suboptimal performance; and 3) w. S2C, the complete method, achieves the best performance-efficiency tradeoff, obtaining optimal performance metrics while significantly reducing parameters and computational cost. Fig. 9 further validates that the spectral curve of w. S2C aligns most closely with the ground truth, significantly outperforming random rewriting and w/o. S2C. This proves that the S2C module can precisely eliminate redundant bands and focus on key spectral information through its adaptive mechanism.

Table VI presents an ablation study on different operation orders in the S2C module. Three variants are compared: A (Ours): upsampling → channel selection → feature refinement; B: channel selection → upsampling → feature refinement; C: channel selection → feature refinement → upsampling. On the Pavia, A achieves PSNR 29.3146, SSIM 0.8127, and SAM 5.3402, outperforming B and C. This corresponds to a PSNR gain of 0.16 dB over B and 0.37 dB over C. This advantage arises because A conducts selection and refinement in the HR space, whereas B operates in the LR space and C upsamples last, limiting HR-aligned processing. In terms of efficiency, A introduces only a minor FLOPs increase while improving performance, and C is cheaper but less accurate. Overall, performing upsampling before spectral selection provides the best tradeoff between performance and computational cost.

TABLE VI
ABLATION STUDY FOR DIFFERENT OPERATION ORDERS IN S2C

Dataset	Variance	FLOPs (G)	PSNR↑	SSIM↑	SAM↓
CAVE	C	11.60	36.3977	0.9148	4.4511
	B	13.48	36.5493	0.9284	4.3798
	A (Ours)	13.58	**36.6905**	**0.9337**	**4.3197**
Pavia	C	12.31	28.9364	0.7933	5.6707
	B	14.20	29.1523	0.8062	5.4511
	A (Ours)	14.29	**29.3146**	**0.8127**	**5.3402**

TABLE VII
ABLATION STUDY ON THE EFFECTIVENESS OF S2M MODULE

Component	Scale	FLOPs	Params	PSNR↑	SAM↓
Downsample	×2	14.79G	4.23M	34.0532	4.1524
w/o. S2M	×2	31.81G	4.08M	34.2531	3.9112
w. S2M	×2	14.19G	4.18M	34.2413	3.9014
Downsample	×4	15.25G	6.74M	29.1441	5.6512
w/o. S2M	×4	32.23G	6.44M	29.3212	5.3521
w. S2M	×4	14.29G	6.54M	29.3146	5.3402

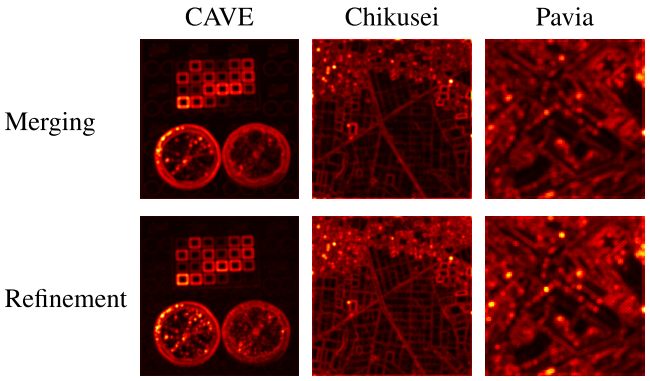


Fig. 10. Gradient magnitude comparison of different datasets after refinement or merging.

2) Ablation of S2M: Table VII presents the ablation results of the S2M module on the Pavia dataset. We compared three configurations: downsample, w/o. S2M, and w. S2M. Simple downsampling leads to significant performance degradation and low efficiency. Removing the S2M module preserves more information but incurs a sharp increase in computational cost (FLOPs). The full S2M module achieves the best performance-efficiency tradeoff: through its semantic-aware token merging mechanism, it dynamically aggregates spectrally similar pixels. While maintaining comparable PSNR and SAM metrics to the computationally intensive "w/o. S2M" configuration, it reduces computational cost by over 50%.

The ablation results for the refinement step are shown in Fig. 10 and Table VIII. In CAVE, refinement enhances object boundaries and grid-like patterns, producing stronger edge responses; in Chikusei, road networks and urban structures become more continuous and spatially coherent; in Pavia, high-intensity responses concentrate near structural boundaries, indicating improved sensitivity to spatial details. Quantitatively, refinement consistently improves PSNR, SSIM,

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TABLE VIII
ABLATION STUDY OF THE REFINEMENT MODULE IN S2M

Dataset	Refinement	×2			×4		
		PSNR↑	SSIM↑	SAM↓	PSNR↑	SSIM↑	SAM↓
CAVE	✗	39.9024	0.9568	3.6912	36.1981	0.9294	4.4111
	✓	40.7917	0.9629	3.4847	36.6905	0.9337	4.3197
Chikusei	✗	43.9368	0.9774	1.9087	38.5716	0.9227	2.8679
	✓	44.6211	0.9803	1.7679	39.6519	0.9361	2.6268
Pavia	✗	33.5689	0.9205	4.3616	28.5468	0.7891	5.6707
	✓	34.2413	0.9393	3.9014	29.3146	0.8127	5.3402

TABLE IX
COMPUTATIONAL RESOURCE REQUIREMENT COMPARISON IN THE INFERENCE STAGE

Methods	FLOPs	Params	FPS	GPU memory
SSPSR	2671.10G	19.13M	11.25n/s	**0.99GB**
MCNet	5275.16G	**1.93M**	2.20n/s	11.58GB
ESSAFormer	851.18G	9.13M	21.94n/s	1.89GB
MambaHSISR	2613.97G	102.65M	1.59n/s	2.49GB
VolFormer	7814.63G	29.82M	0.40n/s	2.55GB
CATANet	229.07G	13.41M	15.59n/s	1.88GB
Ours	**224.67G**	4.30M	**36.87n/s**	1.58GB

and SAM across all datasets. At ×2, PSNR gains are 0.8893 (CAVE), 0.6843 (Chikusei), and 0.6724 (Pavia); at ×4, gains remain notable at 0.4924, 1.0803, and 0.7678, respectively. These results demonstrate that refinement effectively restores information lost during token compression and is essential for optimal spatial and spectral reconstruction.

F. Computational Costs

Under the same hardware and task settings (NVIDIA RTX 4090 GPU, Chikusei dataset, 2× SR), as shown in Table IX, our proposed method demonstrates exceptional inference efficiency. It achieves the best performance in both computational cost (FLOPs, 224.67G) and inference speed (FPS, 36.87 n/s), while its parameter count (4.30M) and GPU memory usage (1.58 GB) are also at very low levels, second only to the MCNet and SSPSR methods, and this is achieved without sacrificing performance. Under the default setting $L = 5$ and 200 epochs, the total training time is approximately 6.5 h. Therefore, our method achieves superior computational efficiency compared to existing lightweight methods like CATANet and performance-oriented methods like Volformer, with almost no loss in reconstruction accuracy, demonstrating significant potential for practical deployment on edge devices.

V. CONCLUSION

This article proposes S^{2}CD-Net to address the challenge of balancing performance and efficiency in hyperspectral image super-resolution on edge devices. Its core is an innovative dual-stage collaborative dimensionality reduction mechanism. Through iterative mutual promotion between dynamic spectral selection and semantic-aware spatial merging, it achieves precise and intelligent reduction of high-dimensional data redundancy. Experiments validate the effectiveness and superiority of the proposed method across multiple dimensions, including quantitative metrics, visual quality, and computational cost. This work not only provides an efficient solution for lightweight HSI-SR but also offers a valuable reference, through its "collaborative dimensionality reduction" concept, for handling other high-dimensional visual data tasks. Future work may integrate this adaptive mechanism with a broader range of low-level vision tasks to further expand its application scope.

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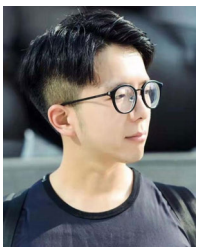
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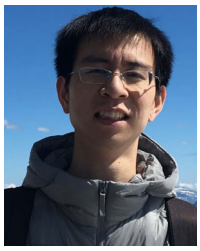
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